

Propylene Glycol Safety: How to Handle Antifreeze Solutions

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Propylene Glycol Safety: How to Handle Antifreeze Solutions We all know how Northern winters can be brutal, with subzero temperatures, icy roads, and frozen industrial plants—but summer is just as unforgiving. In fact, according to NOAA's 130-year climate record, 2024 was the hottest year on record, with an average annual temperature of 55.5°F—3.5 degrees above the 20th-century average. These temperature extremes push our systems to the limit, making glycol management more critical than ever. Glycol isn't just antifreeze, and antifreeze isn't just glycol. Both of them are the backbone of industrial and commercial cooling, keeping industrial operations, manufacturing processes, and specialized equipment running smoothly. But mishandling it comes at a great cost of health risks, environmental contamination, and equipment failure. With mercurial weather straining infrastructure year-round, knowing how to handle propylene glycol toxicity is essential. And at ClearWater Industries, we make sure businesses across Connecticut, Massachusetts, New York, and New Jersey stay ahead of the curve—protecting their operations with expert propylene glycol safety built for every season.

Understanding Glycol and Antifreeze: What They Are and Why They Matter Glycol is a colorless, practically odorless, syrupy liquid that mixes seamlessly with water, making it a go-to solution for temperature regulation in everything from engines to massive HVAC systems. Its primary job is lowering the freezing point and raising the boiling point of water—preventing pipes from bursting in brutal winters and keeping systems from overheating when temperatures climb. There are two main types of glycol used in antifreeze applications:

- Propylene Glycol – The non-toxic, safe chemical, biodegradable option, widely used in food processing, pharmaceutical manufacturing, and commercial HVAC systems. You'll find it in hospitals, schools, and high-rise buildings, where safety and environmental considerations matter.
- Ethylene Glycol – The more efficient alternative but toxic substances found in automotive and industrial applications. Because of its high thermal efficiency, it's commonly used in large-scale cooling systems, such as those in manufacturing plants and commercial properties—where strict handling protocols are essential.

Meanwhile antifreeze—often synonymous with coolant—isn't just about keeping car engines from seizing up. Mixed with water, it circulates through HVAC and plumbing systems, preventing catastrophic failures in sub-zero temperatures. In urban hubs like NYC and Boston, where towering commercial buildings rely on intricate heating and cooling networks, the right antifreeze mix is non-negotiable. Without it, the risk of frozen pipes, costly repairs, and system downtime skyrockets.

Preventative Measures for Industrial Settings: Keeping Glycol Systems in Check A well-maintained glycol system isn't just about topping off fluid but ensuring long-term efficiency, preventing costly damage, and avoiding system failures in extreme weather. Regular monitoring and preventive maintenance keep heating,

cooling, and process systems running reliably. Here's what every facility should be doing: Check Freeze Point & Glycol Concentration Before winter hits, test the glycol percentage using a refractometer or test kit. The right concentration is key to preventing freezing and maintaining optimal heat transfer. If levels drift due to leaks, evaporation, or refills, adjust by adding either glycol concentrate or water to restore the correct freeze protection. Test pH & Inhibitor Levels At least once a year, check the pH of the glycol solution—it should stay between 8 and 10. A dropping pH could signal fluid degradation or corrosion risks. If levels are off, send a sample for a lab analysis to assess inhibitor depletion and glycol breakdown. Restore balance by adding inhibitor buffers from your glycol supplier or, if needed, perform a partial fluid replacement to refresh system integrity. Inspect System Components Corrosion is a silent killer. Look for rust in expansion tanks, discoloration of fluid, or sludge buildup—these are early signs of trouble. Check all air vents since trapped air accelerates corrosion. Pump strainers should be cleaned regularly, as glycol can loosen old scale deposits, leading to sludge buildup that clogs the system. If the system has a glycol feed unit with alarms, verify that low-fluid alerts are operational. Maintain Backup Power & Freeze Safeguards Glycol does its job best when fluid keeps circulating—so backup power for heating loops is critical. Many high-rises in New York City have emergency generators to keep pumps running in a power outage. Schools and commercial buildings should consider fail-safe controls that automatically open zone valves and run pumps if loop temperatures drop too low, preventing freezing in isolated sections. Corrosion Control Challenges in Glycol Systems While glycol itself doesn't cause corrosion, and undergoes a similar process to polyurethane cushions it can still become a major problem if improperly maintained. Over time, inhibitors break down, glycol degrades, and metal surfaces become vulnerable—leading to rust, pitting, leaks, and costly system failures. Facilities in the Northern and Southern US, where glycol is heavily used in high-rises, schools, and commercial properties must stay proactive in corrosion control. Many older buildings still rely on legacy glycol that hasn't been serviced in years, making regular testing and water treatment more essential. Inhibitor Depletion: The Silent Threat When first added, inhibited glycol protects metals by forming a protective barrier. But over time, these inhibitors get “used up” as they neutralize acids and prevent metal degradation. Makeup water (added to replace lost fluid) can further dilute inhibitors, making the glycol less effective. - Warning signs: If pH drops below 8, inhibitors are likely depleted. If it hits 7 or lower, the fluid has turned acidic and is actively corroding system components. - Solution: Annual testing of pH and inhibitor levels is crucial. If depletion is detected, re-inhibiting the fluid or performing a partial fluid replacement can restore protection before damage sets in. Galvanic Corrosion in Mixed-Metal Systems Large building systems often include steel pipes, copper heat exchangers, brass fittings, and even aluminum or galvanized components. If inhibitors aren't properly maintained, glycol can become more conductive than water, accelerating galvanic corrosion between dissimilar metals. - Common issue: Copper-to-steel connections are especially vulnerable. If nitrite inhibitors are depleted, rust can form in steel pipes, while copper ions deposit onto steel, creating galvanic cells that lead to pinhole leaks. - Solution: Ensure glycol contains the right mix of inhibitors (e.g., nitrites for steel, tolyltriazole for copper). Test inhibitor levels every 1–2 years and adjust as needed to prevent unexpected failures. High-Temperature Glycol Breakdown In boilers, heating loops, and rooftop units, glycol can degrade if exposed to excessive heat. Over time, this thermal decomposition produces acidic byproducts that corrode metal components. - Common risk areas: School boiler systems, rooftop HVAC units, or any system with poor circulation or sensor failures that allow temperatures to spike. - Solution: Use industrial-grade glycols with high-temperature stabilizers. Ensure the system doesn't exceed the glycol's rated max temperature, and maintain proper flow to prevent stagnant hot spots. Microbial Growth & Organic Acids While glycol is an organic compound, it can also support bacterial growth if the system lacks proper inhibitors. Certain bacteria can metabolize glycol, producing acidic byproducts that accelerate under-deposit corrosion. - Risk factors: Systems with stagnant warm fluid or those using non-sterile makeup water are prone to microbial contamination. - Solution: Look for cloudiness or sludge in the glycol. Use biocide-treated inhibitors or consider a microbial treatment if

bacteria are detected. Many facility managers treat inhibitor maintenance like adding antifreeze to a car—it's a necessary step to extend system life. You may send a fluid sample for lab testing every 1–2 years to help catch issues early and ensure that the glycol remains protective. Preventing Accidental Exposure and Health Hazards Ethylene glycol exposure poses serious health effects, with acute effects including central nervous system (CNS) severe mental symptoms, metabolic acidosis, and kidney failure, according to the World Health Organization, CDC and NJ Department of Health. Symptoms of inhalation or ingestion can range from dizziness and confusion to severe toxicity requiring emergency medical intervention. Chronic exposure—whether through inhalation exposure of vapors or repeated skin contact—can lead to allergic reactions, neurological damage, organ toxicity, and respiratory irritation. OSHA enforces strict exposure limits, setting the 8-hour time-weighted average (TWA) at 25 ppm and a short-term exposure limit (STEL) at 50 ppm, underscoring the need for safety measures in workplaces handling glycol solutions. To minimize risks, the following countermeasures should be implemented:

- Personal Protection – Workers should wear protective gloves, long sleeves, and safety goggles to prevent direct skin contact and irritant reactions. Ventilation systems and respiratory protection should be in place to reduce vapor inhalation.
- Emergency Response – If glycol contacts skin, wash immediately with soap and water. Ingestion of ethylene glycol is a medical emergency—seek immediate medical attention as untreated exposure can lead to severe organ failure.
- Spill Containment & Safe Storage – Use glycol spill kits for leaks and ensure proper disposal of contaminated materials. Store glycol solutions securely in clearly labeled containers, away from children and pets, as ingestion can be fatal.

Environmental Impact and Proper Disposal of Glycol Improper disposal of glycol-based antifreeze poses serious environmental and health risks, particularly concerning soil and water contamination. According to NJ Health reports, glycol-contaminated soils can become a significant environmental hazard, as pollutants may migrate through rainwater runoff, wind displacement, and physical disturbances. If not properly managed, contaminated soil can spread to adjacent land, surface water sediments, and even groundwater supplies. Additionally, traffic over contaminated areas can generate dust, further distributing glycol residues and contaminants into the surrounding environment. Ethylene glycol, in particular, has a slow degradation rate in water, making it a persistent contaminant if released into storm drains, rivers, or lakes. This can disrupt aquatic ecosystems and pose risks to wildlife. To mitigate these dangers, glycol should never be poured down drains, onto soil, or into water systems. Facilities must follow strict hazardous waste disposal protocols, utilizing certified waste management companies to properly handle used glycol. To reduce environmental impact, many paint and plastics industries implement glycol recycling practices. Through filtration and distillation, glycol can be reclaimed and reused, significantly minimizing waste. Some facilities partner with industrial recycling services to process spent antifreeze into usable products, reducing both environmental harm and disposal costs. By prioritizing proper waste handling and responsible recycling, facilities can limit contamination risks while ensuring regulatory compliance in the disposal of deicing solutions.

Fire and Explosion Risks While antifreeze is not typically flammable, certain conditions can create fire hazards, particularly when glycol vapors become concentrated in poorly ventilated areas. To ensure ethylene and propylene glycol safety, it should have high flash points, meaning they do not easily ignite in liquid form. However, when vaporized at high temperatures or exposed to propylene glycol in open flames, sparks, or hot surfaces, they can catch fire and contribute to combustion. Additionally, mixing glycol with other flammable materials, such as oils or solvents, can increase fire risks. To prevent ignition, always store and handle antifreeze in well-ventilated areas, away from heat sources, smoking, or open flames. In the event of a fire, foam, CO₂, or dry chemical extinguishers should be used—water may be ineffective on glycol fires due to its solubility and potential to spread the material. Immediate evacuation and fire suppression protocols should be followed, ensuring compliance with OSHA and NFPA fire safety standards to maintain a safe working environment.

Regulatory Compliance: Codes and Standards for Propylene Glycol Safety Building and Plumbing Code Compliance Four states: Connecticut, Massachusetts, New York, and New Jersey base their building codes on the International Building Code (IBC) and the

International Mechanical Code (IMC), with state-specific modifications. While the IMC does not prohibit glycol-based solutions, it mandates strict backflow prevention measures to separate glycol from potable water sources. Key Compliance Requirements: - Backflow Prevention: Any make-up water line feeding a glycol loop must be equipped with an approved Reduced Pressure Zone (RPZ) backflow preventer to prevent contamination of the drinking water supply. Plumbing inspectors and health departments closely monitor RPZ installation and require annual testing to confirm functionality. Labeling and Hazard Communication: Pipes and tanks containing glycol solutions must be clearly labeled to prevent accidental cross-connections. - Audits for Older Buildings: Legacy glycol systems, especially in older schools, commercial buildings, and high-rises, should undergo cross-connection audits to ensure compliance with modern safety codes. In some cases, undocumented glycol additions have led to compliance violations that only surface during inspections. Fire Code Compliance: Antifreeze in Sprinkler Systems Historically, glycol-based antifreeze was widely used in fire sprinkler systems to prevent freeze damage in attics, unheated parking garages, and exterior applications. However, after incidents where high concentrations of antifreeze intensified fires, the National Fire Protection Association (NFPA) updated its standards. Key Compliance Requirements (NFPA 13 & NFPA 25): UL-Listed Antifreeze Only: As of September 30, 2022, all antifreeze used in new and existing fire sprinkler systems must be UL-listed. Older systems containing traditional glycerin or propylene glycol mixtures were required to either: - Replace the solution with a UL-listed alternative - Convert the system to a dry-pipe or alternative freeze protection method - Annual Antifreeze Testing: NFPA 25 mandates yearly sampling of sprinkler system antifreeze to verify concentration levels remain within compliant limits. Environmental and Hazardous Material Compliance Glycol disposal and handling regulations fall under state and federal environmental laws, particularly concerning water quality, hazardous waste management and spill prevention. Key Compliance Requirements: State-Specific Waste Regulations: - Connecticut (CT DEEP) classifies used antifreeze under state hazardous waste rules, requiring proper handling and disposal procedures. - Massachusetts (MassDEP) provides guidance on antifreeze recycling and disposal to minimize environmental impact. - EPA Reporting and Spill Containment: While propylene glycol is not federally classified as a hazardous material, large spills may require EPA reporting if they exceed reportable quantities (RQ) or contaminate water sources. - OSHA Workplace Safety Standards: Glycol solutions have OSHA Safety Data Sheets (SDS) that outline proper handling procedures. Workers dealing with concentrated glycol must use appropriate PPE (gloves, eye protection, etc.) to prevent irritation or exposure risks. - Chemical Safety in Schools: School facilities must ensure glycol storage aligns with Healthy School Environment programs, particularly in science labs and maintenance areas where chemical hygiene plans are enforced. Emergency Procedures: Spills, Leaks, and Exposure Response In the event of a spill or leak, it is critical to follow established emergency procedures to mitigate exposure to propylene glycol and environmental contamination. Step 1: Wear Protective Equipment Before handling the spill, put on appropriate PPE, including chemical-resistant gloves, safety goggles, and respirators (if vapors of various chemicals are present) to prevent skin contact, inhalation, or eye irritation. Step 2: Contain the Spill Act immediately to prevent the antifreeze from spreading. Use absorbent materials such as cat litter, sand, or commercial spill absorbents to contain the fluid, especially near drains, water sources, or sensitive areas. Step 3: Neutralize the Spill If applicable, use a neutralizing agent designed for glycol-based solutions to minimize hazards before cleanup. Step 4: Dispose of Waste Properly Collect the absorbed raw material, and contaminated debris in a sealed, labeled hazardous waste container. Follow local and federal hazardous waste disposal regulations to ensure compliance. Step 5: Clean and Decontaminate the Area Use detergent and water to thoroughly wash the spill site, removing any remaining glycol residue. Ensure proper ventilation to disperse vapors and prevent lingering exposure risks. If direct human exposure occurs, immediate treatment and first aid measures are necessary: - Ingestion – Seek emergency medical attention immediately. DO NOT induce vomiting unless directed by medical personnel. Hospital treatment may involve IV antidotes (fomepizole or ethanol) and hemodialysis in severe cases. Ask for certain medicines. - Skin Contact – Remove contaminated clothing

and flush the exposed skin with copious amounts of water for at least 15 minutes. If irritation persists, seek medical evaluation. - Inhalation – Move the individual to fresh air immediately. If breathing difficulties occur, administer oxygen and seek medical attention. Prolonged exposure to vapors, especially in confined spaces, can lead to respiratory arrest and systemic toxicity. Check serum osmolality. By implementing swift containment and proper first aid protocols, facilities can minimize health risks, prevent environmental hazards, and ensure regulatory compliance when handling glycol-based antifreeze solutions. ClearWater's Propylene Glycol Safety Services ClearWater Industries provides a full range of high-quality glycol products tailored for various applications. These products come in a variety of chemical formula to meet the specific needs of different industries: - Safe-T-Therm HD – High-performance glycol with superior heat transfer properties. - Safe-T-Therm AL – Specially designed to prevent corrosion in aluminum-based systems. - Safe-T-Therm GRAS – Food-grade glycol solution meeting stringent safety standards. Comprehensive Service Approach We offer customized system analysis, monitoring, and maintenance, including: - Glycol concentration testing to ensure proper system operation. - Volume studies for accurate system charging. - Laboratory analysis to assess fluid condition and performance. - On-site maintenance to optimize system reliability and efficiency. Each step in our glycol system maintenance process is designed to systematically address and enhance every aspect of your system's performance. Critical Applications We Serve - Industrial Processing Systems – Maintaining temperature stability and preventing downtime. Leading practices in temperature management are crucial to prevent major damages and ensure the longevity of industrial systems. - Commercial Systems – Ensuring freeze protection and optimizing energy efficiency. - Data Center Cooling – Delivering environmental stability for critical IT infrastructure. Partner with ClearWater for Reliable Glycol Solutions Ensuring safe handling and proper maintenance of glycol solutions is essential for maximizing system performance and longevity. In fact, proper glycol system maintenance is crucial for preventing equipment failures. Whether your facility requires glycol system analysis, maintenance, or emergency support, ClearWater Industries provides sustainable water treatment and solutions tailored to your industry needs. Let's protect your investment and prevent costly equipment failures. Contact ClearWater Industries today for a consultation on optimizing your glycol system. To learn more about industrial glycol cooling systems, read here. Frequently Asked Questions (FAQs) What are the safety precautions for handling antifreeze? When handling antifreeze in water treatment process, always wear appropriate Personal Protective Equipment (PPE), including gloves, safety goggles, and protective clothing. Avoid direct skin contact or inhalation of vapors, and store antifreeze in a well-ventilated area away from heat and ignition sources. Ensure proper disposal following local environmental regulations. What are the safety precautions for ethylene glycol? Ethylene glycol is toxic if ingested or absorbed through the skin. It should be stored in clearly labeled, tightly sealed containers away from children and pets. When handling, use OSHA-compliant ventilation systems and protective gear to minimize exposure levels. In case of spills, use appropriate containment and cleanup procedures. What is the proper way to safely handle and store coolants? Coolants should be stored in a cool, dry, and well-ventilated area, away from direct sunlight and incompatible materials. Keep containers sealed when not in use, and always follow manufacturer guidelines. Avoid propylene glycol cross-contamination by using designated storage containers and handling equipment. What are the hazards of glycol? Glycol, particularly ethylene glycol, poses serious human health risks if ingested, inhaled, or absorbed through the skin. It can cause kidney damage, respiratory issues, and neurological symptoms with prolonged exposure. Additionally, glycol spills can contaminate soil and water sources if not properly contained. How should ethylene glycol be stored safely? Store ethylene glycol in approved chemical storage containers, away from extreme temperatures and moisture. Ensure that the storage area has proper spill containment measures in place and is restricted from unauthorized access. Regularly inspect storage tanks for leaks or signs of degradation. What are the safety precautions for cooling systems using glycol? Cooling systems using glycol require regular maintenance, proper dilution ratios, and contamination monitoring to prevent system failures. Operators should conduct routine testing for pH

levels, inhibitor depletion, and signs of corrosion. Always follow manufacturer recommendations for fluid replacement and system flushing. How do you properly handle coolant? Always handle coolant in a well-ventilated area while wearing protective gear. Avoid skin and eye contact, and never mix different types of coolant unless approved by the manufacturer. Dispose of used coolant through a licensed waste disposal facility to prevent environmental contamination. Is antifreeze safe to handle? While antifreeze is safe when handled properly, ethylene glycol-based antifreeze is highly toxic and should be treated with caution. Propylene glycol-based antifreeze is a safer, biodegradable alternative for public use and applications where human and oral exposure is possible. Always follow proper handling and disposal procedures. What is the common name of glycol? Glycol is commonly referred to as ethylene glycol (EG) or propylene glycol (PG), depending on the type. In commercial use, it may also be called coolant, antifreeze, or heat transfer fluid.